

AMAL A K

Full-Stack Software Engineer · MERN · PERN · SaaS · TypeScript · AI Integration
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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with production experience building scalable SaaS applications using MERN and PERN stacks. Architected DiagnomiQ — a multi-tenant healthcare SaaS — with Database-per-Tenant isolation, automated Cloudflare DNS provisioning, JWT-based RBAC, AWS S3 security, and Amazon Bedrock RAG integration. Comfortable owning features end-to-end across REST API design, database architecture, cloud deployment, and React frontends. Experienced in Agile teams across healthcare, e-commerce, and edtech domains.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript

Frontend: React.js, Next.js, Redux, Redux Toolkit, Tailwind CSS, Shadcn/UI, Material UI, Recharts, Vite

Backend: Node.js, Express.js, REST APIs, JWT Authentication, RBAC, Middleware Architecture

Databases & ORMs: PostgreSQL, MongoDB, Prisma ORM, Sequelize, Mongoose, NeonDB

Cloud & DevOps: AWS S3, Amazon Bedrock (RAG), Cloudflare DNS, Vercel, Render, Netlify, Docker (basic), CI/CD

Architecture: Multi-Tenant SaaS, System Design, Microservices, Performance Optimisation, Agile/Scrum, GitHub

PROFESSIONAL EXPERIENCE

Software Engineer | Softnotions Technologies, Technopark, Trivandrum

Jan 2026 – Apr 2026

- Architected DiagnomiQ, a production multi-tenant healthcare SaaS — led full-stack delivery using the PERN stack (PostgreSQL, Express.js, React, Node.js), migrating legacy clinical systems to a scalable cloud platform.
- Designed Database-per-Tenant isolation using PostgreSQL and Sequelize ORM, achieving 100% tenant data separation across dynamically provisioned databases — a strict SaaS security requirement.
- Built fully automated tenant onboarding pipeline: dynamic PostgreSQL database creation, Cloudflare DNS subdomain provisioning per tenant, and AWS S3 folder-level data isolation — reducing manual setup to zero.
- Engineered JWT-based authentication with Role-Based Access Control (RBAC) and custom Tenant DB Middleware, enabling dynamic request routing across isolated per-tenant databases.
- Designed a centralised Platform Database managing tenant registrations, subscription plans, and system-level operations — the core control plane of the multi-tenant SaaS.
- Configured Amazon Bedrock with a RAG pipeline — built a structured medical knowledge base and integrated the AI model via the Bedrock API into the live platform for clinical decision support.

MERN Stack Developer | Relaxplzz Technologies, Technopark, Trivandrum

Apr 2025 – Aug 2025

- Shipped scalable frontend features for GiftoMart, a production e-commerce platform, using React.js, TypeScript, and Tailwind CSS; built RESTful APIs with Node.js and Express.js.
- Developed reusable TypeScript component library, improving cross-team code maintainability; collaborated with product, design, and QA in Agile sprints.

MERN Stack Developer | SciPy Technologies, Trivandrum

Jun 2023 – Jan 2024

- Built RESTful APIs in Node.js and responsive React UIs with Redux state management for a microservices-based ERP system serving an educational institution.
- Managed MongoDB storage with Mongoose schema design, integrated third-party APIs via Axios, and validated endpoint contracts with Postman across Agile sprints.

FEATURED PROJECTS

DiagnomiQ - Multi-Tenant Healthcare SaaS | *React · TypeScript · Node.js · PostgreSQL · Sequelize · AWS S3 · Cloudflare · JWT · Amazon Bedrock*

- Database-per-Tenant architecture with automated PostgreSQL provisioning, Cloudflare subdomain creation per tenant, and AWS S3 folder isolation for HIPAA-aligned medical data separation.
- RAG pipeline on Amazon Bedrock — structured medical knowledge base with Bedrock API integration for AI-powered clinical decision support built into the live SaaS platform.
- Oncology risk dashboards using Recharts — IPI and aalPI survival probability charts, giving oncologists real-time visual risk stratification for lymphoma patient care.

FinSight — Personal Finance Tracker | *Next.js · TypeScript · PostgreSQL · Prisma ORM · Tailwind CSS · Shadcn/UI · Vercel*

- Type-safe full-stack finance tracker with Prisma ORM + PostgreSQL; interactive analytics dashboards with category filtering — deployed on Vercel with edge optimisation for global performance.

GiftoMart — Gift Management Platform | *React · Node.js · Express.js · MongoDB · TypeScript · Mongoose*

- Gift creation and order-tracking modules with JWT authentication and reusable TypeScript components; REST APIs for product and user services using Node.js, MongoDB, and Mongoose ODM.

EDUCATION

B.Sc. Computer Science | Emmanuel College, Vazhichal

2018 – 2021